

Coronary centerline tortuosity: choices and mathematics

Working document, ImageCAS-X development cohort

26 September 2026 (provisional, not pre-registered)

Status. Every choice below was made on the ImageCAS-X training split (560 scans), which is a development cohort: clinically referred patients, about half with plaque, mixed cardiac phase, no age or sex. Nothing here is a normal range. No descriptor has yet been tested for agreement between reference and CAS-Net-derived centerlines, and the region limits were chosen on the same scans they are evaluated on. They must be frozen in a dated commit and checked on the validation split before any thesis claim.

1 Input and preprocessing

- **Data.** Only the delivered centerlines, `<id>.coronary_{left,right}_centerline.vtk`: LPS coordinates in mm, per-point `segment_label`, native spacing about 0.45 mm.
- **Vessel path.** For a trunk label t (RCA 9, LAD 2, LCx 3) the path is the longest geodesic path through edges whose two ends both carry label t . It is oriented from the ostium (RCA, via the `start_points` array) or from the LM (LAD, LCx) to the distal end.
- **No extra smoothing.** The dataset regenerated the centerlines by skeletonising the corrected masks and applying a Gaussian of $\sigma = 0.5$ mm over a 5-vertex window. The path is only resampled at $h = 0.25$ mm of arc length, giving points $\mathbf{x}_0, \dots, \mathbf{x}_n$ with arc length $s_i = \sum_{j < i} \|\mathbf{x}_{j+1} - \mathbf{x}_j\|$ and $L = s_n$. Adding $\sigma = 1$ mm changed κ_a medians by 2 to 4 % (Spearman 0.98 to 0.99). Any centerline from another source (CAS-Net, CGPS) must first pass through the same skeleton plus $\sigma = 0.5$ mm step.
- **Exception.** Descriptors that need a second derivative (Section 2.4) use an extra Gaussian of $\sigma_0 = 1.25$ mm along arc length, fixed from a noise bound before any cohort number.

2 Descriptors

2.1 Arc/chord (distance metric)

$$\text{DM} = \frac{L}{\|\mathbf{x}_n - \mathbf{x}_0\|}, \quad \text{DM}_{2D} = \frac{\sum_i \|P(\mathbf{x}_{i+1} - \mathbf{x}_i)\|}{\|P(\mathbf{x}_n - \mathbf{x}_0)\|},$$

where $P \in \mathbb{R}^{2 \times 3}$ projects onto the image plane of an angiographic view. The viewing direction for LAO/RAO angle α and cranial/caudal angle γ is $\mathbf{d} = (\sin \alpha \cos \gamma, -\cos \alpha \cos \gamma, \sin \gamma)$ in LPS, and P has rows $\mathbf{e}_1 = \mathbf{d} \times \hat{\mathbf{z}} / \|\cdot\|$ and $\mathbf{e}_2 = \mathbf{d} \times \mathbf{e}_1$. Standard views: RCA LAO 30 ($\alpha = 30^\circ$), LAD RAO 20/CRA 30, LCx RAO 20/CAU 30. DM measures the overall course and grows with vessel length (Spearman 0.70 to 0.84 per whole vessel).

2.2 Mean chord curvature κ_a (primary)

Chord resampling places vertices $\mathbf{v}_0 = \mathbf{x}_0, \mathbf{v}_1, \dots$ on the curve with $\|\mathbf{v}_{k+1} - \mathbf{v}_k\| = \ell$ exactly (solved on each polyline segment). With unit chords $\mathbf{u}_k = (\mathbf{v}_{k+1} - \mathbf{v}_k) / \ell$ and turning angles

$\theta_k = \arccos(\mathbf{u}_{k-1} \cdot \mathbf{u}_k)$ at vertex $\mathbf{v}_k$,

$$\kappa_a^{(\ell)} = \frac{1}{L} \sum_k \theta_k \quad [\text{rad/mm}], \quad \ell = 5 \text{ mm}.$$

Total turning is $\Theta = \sum_k \theta_k = \kappa_a L$. For a smooth curve $\theta_k \approx \int_{\text{chord}} \kappa ds$, so κ_a is a low-pass mean absolute curvature at scale ℓ . It is insensitive to vessel length (Spearman 0.2 to 0.3), not affected by dominance or image quality, and it cannot see a bend closer than one chord to the path end (so it ignores an ostial hook by construction).

2.3 Bends and the largest bend angle

At $\ell = 1$ mm chords, a vertex is *bending* if $\theta_k > \kappa_{\min} \ell$ with $\kappa_{\min} = 0.02 \text{ mm}^{-1}$ (bend radius under 50 mm). A bend is a maximal run of bending vertices, split where the binormal $\mathbf{b}_k = \mathbf{u}_{k-1} \times \mathbf{u}_k$ reverses ($\mathbf{b}_k \cdot \mathbf{b}_{k-1} < 0$). Its angle is $\sum_{k \in \text{run}} \theta_k$ and its position the θ -weighted mean arc length. Reported: the **largest bend angle** per region. Bend counts ($\geq 45^\circ$, $\geq 90^\circ$) are descriptive only, because they depend on $\kappa_{\min}$ and smoothing.

2.4 Bishop curvature spectrum

On the σ_0 -smoothed curve $\mathbf{c}(s)$ with unit tangent $\mathbf{T}$, carry a rotation-minimising (Bishop) frame $(\mathbf{T}, \mathbf{M}_1, \mathbf{M}_2)$ by parallel transport and write the curvature vector $\mathbf{T}'$ as one complex number

$$\psi(s) = k_1 + ik_2 = \mathbf{T}' \cdot \mathbf{M}_1 + i \mathbf{T}' \cdot \mathbf{M}_2, \quad |\psi| = \kappa.$$

ψ needs no torsion, is defined through inflections, and for a planar curve is the signed 2D curvature up to a constant phase. This is what unifies the 2D and 3D descriptions.

- **Bending energy.** $k_B = \left(\frac{1}{|R|} \int_R |\psi|^2 ds \right)^{1/2}$ (mm^{-1}), the RMS curvature over region R .
- **Out-of-plane share.** With a Gaussian window g_W of standard deviation $W = 5$ mm,

$$f_{\text{twist}} = 1 - \frac{\int_R |g_W * \psi^2| ds}{\int_R g_W * |\psi|^2 ds} \in [0, 1].$$

Squaring removes the sign, so $g_W * \psi^2$ is large only when bending keeps one plane over the window. $f_{\text{twist}} = 0$ exactly for any planar curve, whatever its inflections; for a helix of curvature κ and torsion τ it is $1 - e^{-2\tau^2 W^2}$.

- **Wiggle share** (dropped, defined for completeness). An orthonormal DCT of ψ is split by a power-complementary Gaussian, $|H_L(\omega)|^2 = e^{-(\omega s_c)^2}$, $|H_H|^2 = 1 - |H_L|^2$, $s_c = \sqrt{\ln 2} \lambda_c / 2\pi$, $\lambda_c = 20$ mm; $f_{\text{wiggle}} = \int_R |\psi_H|^2 / \int_R (|\psi_L|^2 + |\psi_H|^2)$.

The classical indices are moments of the same ψ spectrum: RMS curvature is moment 0, a Grisan-type turn index is about moment -1 , and arc/chord is about moment -2 , which explains why arc/chord follows length and truncation.

2.5 Regional attribution

Every quantity is computed once on the whole vessel and then attributed to a region $R = [a, b)$ of arc length: $\kappa_a(R) = \sum_{k: s(\mathbf{v}_k) \in R} \theta_k / |R|$; bends are assigned by their centre; k_B , f_{twist} and f_{wiggle} integrate their pointwise densities over R ; DM uses the sub-curve. Regional totals therefore add up to the whole-vessel value and no region loses its edges to a chord or a filter. Regions shorter than 10 mm, or without their landmark, give an explicit NaN row with the reason.

3 Regions per vessel

Take-off positions are the arc length of the trunk point joined by an edge to the first point of the side branch label (nearest trunk point within 3 mm if no edge exists).

Vessel	Proximal	Mid	Distal / end window
RCA	5 to 40 mm (first 5 mm trimmed)	40 mm to $L - 20$	crux: last 20 mm before R-PDA/R-PLA
LAD	LM to D1	D1 to D2, else halfway to end	rest
LCx	LM to OM1	OM1 to OM2, else halfway to end	rest

- **RCA 5 mm trim.** Many RCAs start with a sharp hook where the centerline leaves the aortic root. In scan 662 it holds most of the vessel’s bending energy; trimming 5 mm cut the proximal k_B from 0.121 to 0.041 mm⁻¹. The LAD and LCx show no such hook (whole vs trimmed κ_a , Spearman 0.99 to 1.00), so they are not trimmed.
- **Crux window.** Defined only when the RCA label ends at R-PDA/R-PLA: 512 of 513 right-dominant, 6 of 15 co-dominant, 0 of 32 left-dominant scans.
- **Landmark coverage.** Measurable proximal regions: RCA 560, LAD 424 (D1 median 18 mm from the LM, 127 regions under 10 mm), LCx 365 (OM1 missing in 69 right-dominant scans, 126 under 10 mm). Recommendation (untested for LAD and LCx): fixed proximal windows as primary, landmark segments as secondary.

4 Evidence behind the choices (560 training scans)

Regional κ_a values are nearly independent, so a single whole-vessel number hides where bending sits (Spearman between regions):

	RCA (prox _{trim} /mid/crux)	LAD (prox/mid/distal)	LCx (prox/mid/distal)
proximal vs mid	0.55	0.13	0.14
proximal vs distal/crux	0.21	0.20	0.10
mid vs distal/crux	0.24	0.37	0.25
whole vs distal/crux	0.43	0.89	0.71

The whole-LAD value is mostly the distal (apical) LAD, which is not where first plaque forms.

Added information. Share of each descriptor’s rank variance *not* explained by (κ_a , vessel length), in the plaque-relevant proximal and mid regions:

Descriptor	RCA	LAD	LCx
f_{twist}	94 to 96 %	98 to 99 %	98 to 99 %
largest bend angle	39 to 57 %	69 to 83 %	66 to 70 %
regional DM (3D)	19 to 43 %	64 to 66 %	44 to 60 %
k_B	19 to 30 %	48 to 69 %	41 %
mean 10 mm arc/chord	16 to 17 %	46 to 68 %	42 %

Mean 10 mm arc/chord and k_B agree with each other at Spearman 0.88 to 0.98 in every proximal and mid region, so at most one of them can be kept. In short regions some of the apparent new information may be estimation noise.

Confounds. Proximal and mid κ_a : no dominance effect (all $p \geq 0.2$), $|\rho| \leq 0.09$ with image quality. Whole-vessel DM shifts with dominance in all three vessels (p from 10⁻⁵ to 10⁻¹⁰). Distal LCx descriptors follow length (κ_a ρ 0.47, bends 0.61), because the distal LCx extent depends on dominance.

5 Decision

Status	Descriptor	Where / condition
Primary	$\kappa_a^{(5)}$	every plaque-relevant region
Secondary	largest bend angle	same regions
Conditional	f_{twist}	proximal and mid, only if reliable
Literature only	whole-vessel DM (3D and 2D view), whole-vessel κ_a	reported with length and dominance
Descriptive	bend counts $\geq 45^\circ$; RCA crux; distal LCx	distal LCx stratified by dominance
Exploratory	k_B	RCA only after the 5 mm trim
Dropped	mean 10 mm arc/chord, total turning, f_{wiggly} , regional 2D DM, regional bends $\geq 90^\circ$, inflection count, ICM, full SOAM, writhe, box-counting dimension, SRVF distance to a line	

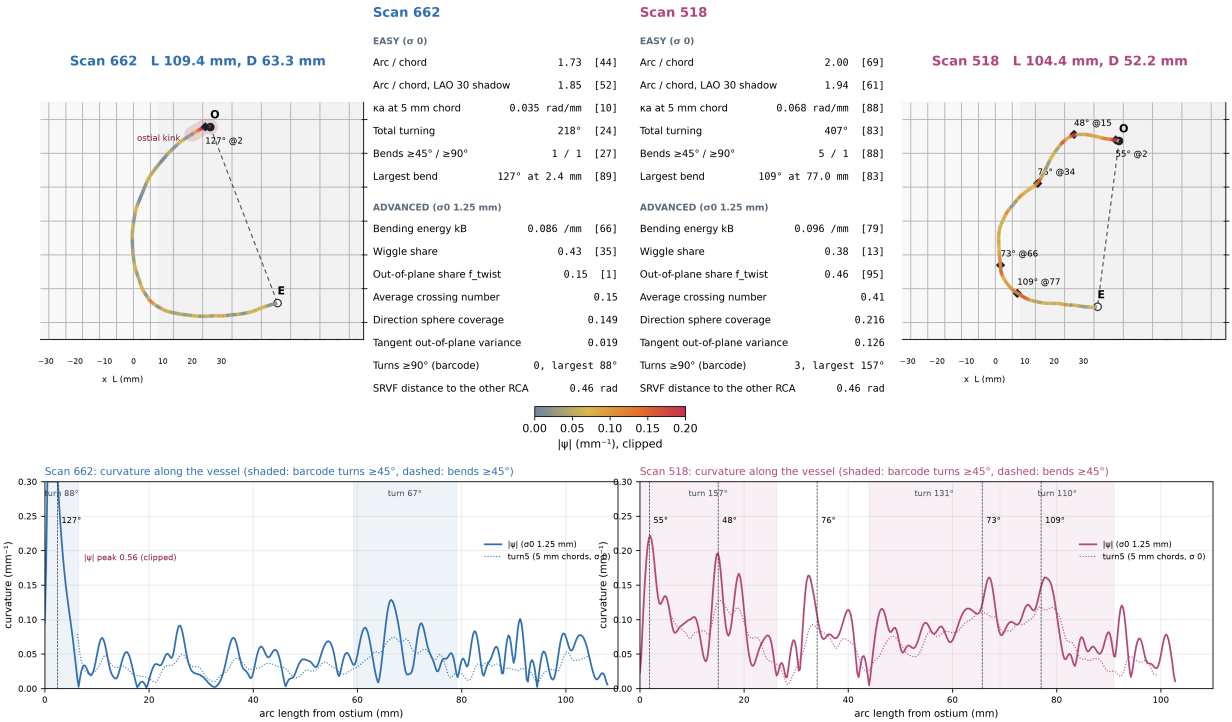
Open before adoption.

1. Reliability per region: jitter and, above all, reference versus CAS-Net-derived centerlines. Whole-vessel f_{twist} fell to ICC 0.15 to 0.32 under 0.5 mm correlated jitter, while κ_a held 0.74 to 0.81.
2. Fixed-window versus landmark proximal regions for the LAD and LCx.
3. Freeze regions and thresholds in a dated commit; confirm on the validation split.
4. CGPS needs a labeller (no artery names) and the same smoothing step before any of this applies.

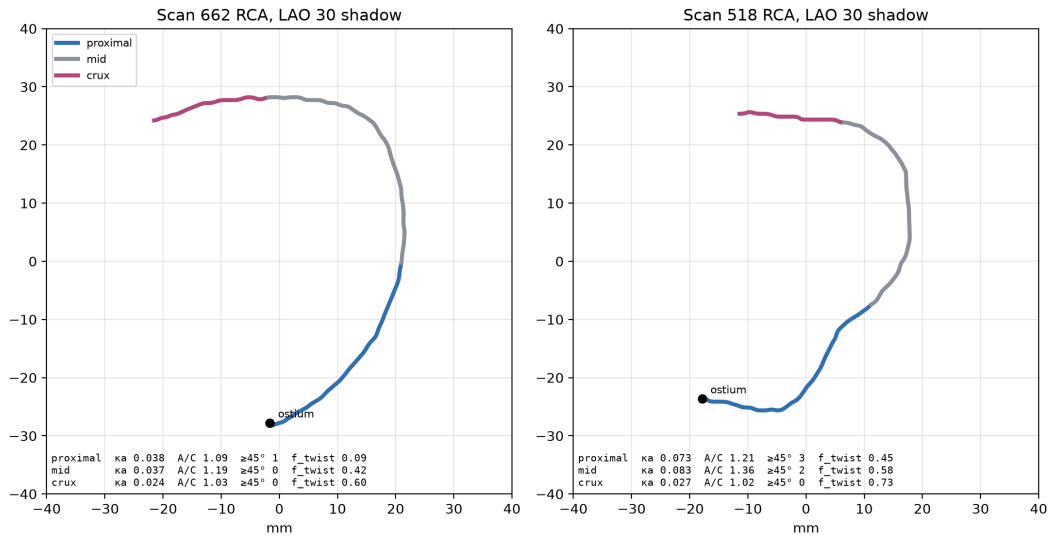
Figures

RCA tortuosity comparison: scan 662 (smooth) vs scan 518 (tortuous), LAO 30 orthographic, same mm scale

Colour = $|\kappa|$ curvature (mm^{-1}), shared scale clipped at 0.2 (radius 5 mm). Dots: ostium (O), distal end (E), bends $\geq 45^\circ$ with angle. Dashed: chord. Grid 10 mm, superior up. Percentiles among 560 training RCAs in brackets.



Two length-matched RCAs (scans 662, 518) in LAO 30 with easy and advanced descriptors.



The same RCAs split into proximal, mid and crux regions (before the 5 mm trim is applied).

Code and outputs

Scripts in `tortuosity/research/`: `simple/compute_simple.py`, `unified/bishop.py`, `case_study/easy.py`, `advanced.py`, `regional.py`, `regional_analysis.py`, `visualise.py`. Outputs in `/work3/s254124/imagecasx_results/tortuosity_research/case_study/`: `regional_train560.csv`, `regional_lad_train560.csv`, `regional_lcx_train560.csv` and the matching `_analysis.txt` files. Rebuild with `latexmk-pdf` `tortuosity_methods.tex`.